

ONLINE BANKING **TRANSACTION SYSTEM**

Online Banking Transaction System
Software Requirements Specification
Version <1.0>

Online Banking Transaction System	Version: <1.0>
Software Requirements Specification	26/08/2026

Revision History

Date	Version	Description	Author
23/08/2026	1.0	SRS 1.0	Group-3

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Software Requirements Specifications

1. Introduction

The introduction provides an overview of the Software Requirements Specification (SRS) document for the Online Banking Transaction System. It outlines the purpose, scope, definitions, references, and an overview of the system to ensure a clear understanding of the project requirements and objectives. The detailed requirements of the Online Banking Transaction System are provided in this document.

1.1 Purpose

The purpose of this document is to define the requirements for the Online Banking Transaction System. This SRS will serve as a guide for the development, testing, and maintenance of the system. It describes the functionalities, performance, and design constraints necessary for the successful implementation of the system.

The system is intended to simplify the process of accessing bank accounts and performing online banking transactions. It provides customers with a centralized platform to log in securely, view account information, check account balance, transfer funds, and view transaction details. Bank administrators can manage customer information and monitor banking transactions.

1.2 Scope

The scope of the Online Banking Transaction System includes managing customer accounts, account balances, fund transfers, transaction records, and transaction status. The system is designed to streamline and automate online banking activities, providing efficient and secure management of banking transactions.

The system will provide customers with facilities to securely log in, view their account balance, transfer funds, and view transaction history. Bank administrators will be able to manage customer and account information and monitor transactions accordingly. The system will be developed using Spring Boot for the backend and MySQL for storing customer, account, and transaction-related information.

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1.3 Definitions, Acronyms, and Abbreviations

OBTS	Online Banking Transaction System
SRS	Software Requirements Specification
GUI	Graphical User Interface
Admin	Administrator
MVC	Model-View-Controller
API	Application Programming Interface
OTP	One-Time Password

1.4 References

- OBTS Structural Model
- OBTS Behavioral Model
- OBTS NFR Model
- Vision Draft for OBTS

1.5 Overview

The document is structured to provide a comprehensive description of the Online Banking Transaction System, including user characteristics, hardware and software requirements, and specific functionalities. Section 2 gives an overall description, while Section 3 details the specific requirements. Section 4 provides supporting information for further reference.

2. Overall Description

This document addresses the current challenges faced by existing banking processes, which may make it difficult to securely manage customer accounts, account balances, fund transfers, transaction records, and transaction statuses. It outlines the key stakeholders and users of the proposed Online Banking Transaction System, capturing their needs and expectations.

The document also provides a summary of the primary features and a brief description of the proposed system. The Online Banking Transaction System provides a centralized platform where customers can securely access their accounts, check balances, transfer funds, and track their transaction history. Bank administrators can manage customer and account information and monitor banking transactions.

The following Software Requirements Specification (SRS) offers a detailed product perspective from various stakeholders involved. It outlines the comprehensive functions of the Online Banking Transaction System, including user characteristics, system constraints, assumptions, dependencies, and the subsets of requirements necessary to achieve the system's goals.

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3. Specific Requirements

The specific requirements for the Online Banking Transaction System are categorized under functionality, usability, reliability, performance, security, supportability, and design constraints.

3.1 Functionality

Introduction –

This subsection contains the requirements for the Online Banking Transaction System, organized by the features discussed in the vision document. Features from the vision document are then refined into use case diagrams and sequence diagrams to best capture the functional requirements of the system. All these functional requirements can be traced using traceability matrices, ensuring that the system meets the specified needs and objectives.

3.1.1 User Management

- The system shall allow the management of customer information.
- The system shall authenticate users during login.
- The system shall allow users to access the system based on their respective roles.
- The system shall maintain information about employees, administrators, or managers.
- The system shall provide secure access to customer accounts.

3.1.2 Account Management

- The system shall allow customers to view their account information.
- The system shall allow customers to check their account balance.
- The system shall store and maintain customer account information.
- The system shall display available account balance to the customer.
- The system shall support Savings and Current account types.
- The system shall validate that a Savings account balance does not fall below the minimum required balance after a debit transaction.
- The system shall validate that a Current account does not exceed its permitted overdraft limit after a debit transaction.
- The system shall verify that both source and destination accounts are active before processing any transaction.
- The system shall maintain records of customer accounts.

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3.1.3 Fund Transfer Management

- The system shall allow customers to initiate fund transfers.
- The system shall allow customers to provide the required details while submitting a fund transfer.
- The system shall validate the required transaction details.
- The system shall maintain records of fund transfer requests.
- The system shall enforce a daily maximum transfer limit for each account.
- The system shall track the cumulative amount transferred by an account within a day.
- The system shall reject fund transfer requests that exceed the daily transaction limit.
- The system shall process fund transfer requests.

3.1.4 Transaction Management

- The system shall allow customers to perform online banking transactions.
- The system shall store transaction information.
- The system shall maintain records of completed and unsuccessful transactions.
- The system shall update account balances based on successful transactions.
- The system shall allow customers to view their transaction history.
- The system shall refund the debited amount to the source account if crediting the destination account fails after a successful debit.
- The system shall notify customers upon the successful completion of a transaction.
- The system shall notify customers if a transaction fails or is rejected, along with the reason whenever available.
- The system shall maintain transaction consistency by reversing partially completed transactions whenever possible.
- The system shall escalate a transaction for manual intervention if the automated refund process fails.
- The system shall maintain transaction details for future reference.

3.1.5 Transaction Status Tracking

- The system shall allow customers to check the status of their transactions.
- The system shall track and display transaction status as Pending, Validating, Debiting, Crediting, Success, Failed, Compensated, or Escalated.
- The system shall update the status of a transaction after it has been processed.
- The system shall allow customers to view the updated status of their transactions.

3.2 Usability

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- The system shall provide a user-friendly interface with consistent navigation across all pages.
- The system shall provide clear and understandable functionalities for customers and administrators.
- The system shall provide clear forms for performing banking transactions.
- The system shall clearly display account balances and transaction statuses.
- The system shall clearly display account balances and transaction statuses.
- The system shall provide clear messages for successful and unsuccessful transactions.
- The system shall minimize the number of steps required to complete common banking tasks.

3.3 Reliability & Availability

- The system shall store customer, account, and transaction-related information reliably.
- The system shall maintain consistent records of accounts and transactions.
- The system shall regularly back up system data to prevent data loss.
- The system shall handle invalid or incomplete input appropriately.
- The system shall handle concurrent access to account balances to prevent race conditions during simultaneous transactions.
- The system shall ensure that account balances remain consistent when multiple transactions are processed at the same time.
- The system shall ensure the availability of the system to authorized users.

3.4 Performance

- The system shall respond to user actions within a reasonable time under normal load conditions.
- The system shall efficiently process online banking transactions.
- The system shall efficiently retrieve customer, account, and transaction-related information.
- The system shall support multiple users accessing the system.
- The system shall recover gracefully from network or destination account failures during fund transfers and maintain transaction consistency.
- The system shall scale to accommodate future growth.

3.5 Security

- The system shall authenticate users before providing access to the system.
- The system shall restrict access based on user roles.
- The system shall protect customer, account, and transaction-related information.

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- The system shall securely store user credentials and sensitive information.
- The system shall validate user input to reduce invalid data entry.
- The system shall ensure that only authorized users can perform banking transactions.
- The system shall use appropriate authentication mechanisms for sensitive transactions.
- The system shall reject transactions involving frozen, inactive, or closed accounts.
- The system shall allow transactions only between accounts that are verified as active.

3.6 Supportability

- The system shall provide appropriate documentation for users and administrators.
- The system shall support regular system updates and maintenance.
- The system shall provide appropriate error handling and debugging support.
- The system shall be designed to allow future enhancements and modifications.

3.7 Design Constraints

- The system shall be developed using Spring Boot for backend development.
- The system shall use MySQL for storing customer, account, and transaction-related information.
- The system shall follow a Layered Architecture.
- The system shall use the MVC design pattern where applicable.
- The system shall use the Singleton design pattern where required.
- The system APIs shall be tested using Postman.
- The system development shall consider time, resource, and project constraints.

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3.8 On-line User Documentation and Help System Requirements

- The system shall provide appropriate guidance for using the main functionalities.
- The system shall provide information on how to access an online banking account.
- The system shall provide information regarding account balances and transaction history.
- The system shall provide guidance for customers to perform fund transfers.
- The system shall provide appropriate error messages and instructions where necessary.

3.9 Purchased Components

- Not Applicable

3.10 Interfaces

3.10.1 User Interfaces

- The system's user interface shall provide clear and consistent navigation.
- The system shall provide appropriate interfaces for customers and administrators.
- The system shall use appropriate visual elements and typography.
- The system interface shall clearly display account information, account balances, transactions, and transaction statuses.

3.10.2 Hardware Interfaces

- The system shall operate on standard computing devices.
- The system shall specify hardware requirements such as servers, storage, and network equipment where required.
- No specialized hardware interfaces are required for the core functionality of the system.

3.10.3 Software Interfaces

- The system shall use Spring Boot as the backend development framework.
- The system shall use MySQL as the database management system.
- The system shall provide APIs that can be tested using Postman.
- The system shall support communication between the application and the MySQL database.

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- The system shall ensure compatibility between the software components used in the application.

3.10.4 Communications Interfaces

- The system shall use HTTP/HTTPS for communication over the internet.

3.11 Licensing Requirements

- Not Applicable

3.12 Legal, Copyright, and Other Notices

- The system shall include appropriate legal and copyright notices.
- The system shall handle employee-related information responsibly.
- The system shall comply with relevant organizational policies and regulations.

3.13 Applicable Standards

- The system shall comply with relevant data protection and privacy standards.
- The system shall identify and comply with applicable industry standards.
- The system shall follow appropriate software quality and security practices.

4. Supporting Information

Please refer to the following documents:

1. Vision document for OBTS.
2. Use case analysis.
3. Structural models.
4. Behavioral models.
5. Non-functional requirements model.
6. Traceability Matrix.
7. Project Plan

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